M4: Short Answers on Rapidity

Complete Solutions with Clarified Questions, Teaching, Analysis & Derivations

PHYS 7502 — Prof. Frantz

Reference Definitions (from Class Slides)

Key Concept

Class (Frantz) definition of rapidity y: defined with respect to the *beam axis* (z-direction), using the longitudinal component $p_z \equiv p_{\parallel}$:

$$\boxed{y = \frac{1}{2}\ln\left(\frac{E + p_z}{E - p_z}\right) = \frac{1}{2}\ln\left(\frac{1 + \beta_z}{1 - \beta_z}\right) = \operatorname{arctanh}(\beta_z)} \tag{1}$$

where $\beta_z = p_z/E = \beta\cos\theta$, with θ the polar angle from the beam.

Pseudorapidity η: a purely geometric/angular quantity (no mass or energy needed):

$$\boxed{\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right] = \frac{1}{2}\ln\left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}\right) = \operatorname{arctanh}\left(\frac{p_z}{|\vec{p}|}\right)} \tag{2}$$

Textbook definition (e.g. Mann Eq. 2.6 p. 41; L&B Eq. 9.43): rapidity of the *Lorentz boost itself*:

$$\gamma = \cosh y, \qquad \beta\gamma = \sinh y, \qquad \beta = \tanh y \tag{3}$$

Part (a)

Original Question (verbatim)

In all cases in this problem when referring to rapidity y, the definition given in the class slides is meant, rather than e.g. the one defined by formulas connecting the boost factors to hyperbolic trig functions such as $\gamma = \cosh y$ found in many books such as Mann (problem #2, 2.6 pg ˜41) or L&B (˜Eq 9.43). What's the difference between our definition and that textbook definition? (Note, the Thomson other alt textbook, Modern Particle Physics, does include a short treatment on the rapidity definition used in my slides, in section 10.9.1).

What This Is Really Asking (Plain English)

Two different formulas both go by the name "rapidity." How do they differ?

Formula A (our class / collider physics): $y = \operatorname{arctanh}(\beta \cos\theta) = \frac{1}{2}\ln\left(\frac{E+p_z}{E-p_z}\right)$
Formula B (most textbooks, e.g. Mann, L&B): $y = \operatorname{arctanh}(\beta)$ i.e. $\gamma = \cosh y$, $\beta = \tanh y$

State what is physically different between these two definitions and under what special condition they become equal.

Teaching & Derivation

The textbook definition (3) treats rapidity as a property of a *one-dimensional Lorentz boost*. It parametrizes the full boost velocity β via $\beta = \tanh y$, so $\gamma = \cosh y$. This implicitly assumes the particle's momentum is *entirely along the boost direction* (i.e. $\theta = 0$ or $180°$, meaning $\vec{p}$ is parallel to $\hat{z}$).
In contrast, the class definition (1) is for a particle moving at *any angle* θ relative to the beam. It uses only the *longitudinal projection*: $\beta_z = \beta\cos\theta = p_z/E$. Therefore:

$$\tanh y_{\text{class}} = \beta\cos\theta = \beta_z \qquad \text{vs.} \qquad \tanh y_{\text{text}} = \beta$$

These are equal **only when** $\cos\theta = 1$, i.e. $\theta = 0°$ or $180°$ (particle moving exactly along the beam axis, so $p_T = 0$).

Physical picture: The class definition is tailored for collider physics, where we decompose every particle's momentum into p_z (along the beam) and p_T (transverse). It assigns a "longitudinal rapidity" to a particle at any angle. The textbook definition describes the rapidity of a reference *frame* or a particle in 1D motion.

Connection via transverse mass: With $m_T = \sqrt{m^2 + p_T^2}$, our definition gives $E = m_T\cosh y$, $p_z = m_T \sinh y$. The textbook gives $E = m\cosh y$, $|\vec{p}| = m\sinh y$. These coincide only when $p_T = 0$ (so $m_T = m$ and $p_z = |\vec{p}|$).

Answer (2 sentences max per sub-question)

The class definition $y = \operatorname{arctanh}(\beta \cos\theta)$ uses only the *longitudinal* (beam-direction) component of the velocity, so it applies to particles at any angle θ relative to the beam. The textbook definition $\gamma = \cosh y$, $\beta = \tanh y$ parametrizes the particle's *full* speed and agrees with ours only when the particle travels exactly along the beam axis ($\theta = 0^\circ$ or 180°, i.e. $p_T = 0$).

Part (b)

Original Question (verbatim)

For practical purposes (in most cases where they are used) the numerical values of either pseudorapidity η or rapidity y, compared for the same particle are best described as which of the following: very different, different, similar, effectively the same, almost exactly the same, or exactly the same—include brief justification of this, though you may simply refer to information given in class. Which is preferred when considering when and if a maximum value should be considered?

What This Is Really Asking (Plain English)

This question has two sub-parts:

(b.1) If you compute both η and y for the *same* particle at a typical collider experiment, how close are the two numbers? Pick the best descriptor and justify briefly:

very different / different / similar / effectively the same / almost exactly the same / exactly the same

(b.2) η can go to $\pm\infty$ (as $\theta \to 0°$ or $180°$), but y for the beam particle is finite. Which variable (y or η) is more meaningful when asking "what is the maximum relevant value?" and why?

Teaching & Derivation

Compare the two definitions side by side:

$$y = \operatorname{arctanh}\left(\frac{p_z}{E}\right) \qquad \eta = \operatorname{arctanh}\left(\frac{p_z}{|\vec{p}|}\right) \tag{4}$$

The **only** structural difference: y has energy E where η has $|\vec{p}|$. Since $E = \sqrt{|\vec{p}|^2 + m^2} \geq |\vec{p}|$, with equality when $m = 0$.

Consequences:

- **Massless particles** (γ, ν, effectively e): $E = |\vec{p}|$ exactly $\Rightarrow$ $y = \eta$ **exactly**.
- **Massive particles at high energy** ($|\vec{p}| \gg m$): $E \approx |\vec{p}|$ $\Rightarrow$ $y \approx \eta$ to excellent precision, except at very forward/backward angles.
- At midrapidity ($\theta \approx 90°$), even for protons at collider energies, $|y - \eta|$ is negligibly small.

On the maximum value: Pseudorapidity of the beam $\eta_{\text{beam}} = -\ln[\tan(0/2)] = \infty$ (formally, since $\theta = 0$). But the beam *rapidity* is finite: $y_{\text{beam}} = \operatorname{arctanh}(\beta_{\text{beam}})$. Since all produced particles have $|y| \lesssim y_{\text{beam}}$, rapidity y provides a natural, finite upper bound. Pseudorapidity has no such finite limit, so y is preferred for bounding the kinematic range.

Answer (2 sentences max per sub-question)

For practical purposes η and y are best described as **effectively the same** (or "almost exactly the same"), because at collider energies $E \approx |\vec{p}|$ for most detected particles; the only structural difference (E vs. $|\vec{p}|$) vanishes as $m/p \to 0$. Rapidity y is preferred when considering a maximum value: the beam rapidity $y_b = \text{arctanh}(\beta_{\text{beam}})$ is finite and physically meaningful, while the beam pseudorapidity $\eta_b \to \infty$ (since $\theta_{\text{beam}} = 0$).

Part (c)

Original Question (verbatim)

Give two advantages of using y or η rather than the polar angle θ. Hint: relate at least one to the distribution of particles as a function of any of the three variables. For practical purposes, do these advantages apply to both y and η, or just one or the other?

What This Is Really Asking (Plain English)

Name two concrete reasons why experimentalists plot data vs. y or η instead of the polar angle θ.

(c.1) What is advantage #1? (Hint: think about what happens to particle distributions when you Lorentz-boost along the beam.)
(c.2) What is advantage #2? (Hint: sketch or compare $dN/d\theta$ vs. dN/dy—which looks more "usable"?)
(c.3) For each advantage, state whether it applies to y only, η only, or both.

Teaching & Derivation

Advantage 1 — Lorentz-invariant differences (additivity under boosts): Under a longitudinal Lorentz boost with velocity β_F:

$$y' = y + y_{\beta_F}, \qquad \text{so} \qquad \Delta y = y_2 - y_1 = y_2' - y_1' \quad \text{(invariant!)}$$

Distributions dN/dy have the **same shape** in every boosted frame—merely shifted by a constant. This is enormously useful for comparing fixed-target and collider experiments, and for defining boost-invariant jet sizes ($\Delta R = \sqrt{\Delta y^2 + \Delta\phi^2}$).
The angle θ, by contrast, transforms via relativistic aberration—a complicated, energy-dependent formula.

Advantage 2 — Flat particle distributions: At high-energy colliders, $dN/d\theta$ is extremely peaked near $\theta \approx 0°$ and 180° (most particles fly close to the beam). Plotting dN/dy or $dN/d\eta$ "unfolds" this peaking into a broad, roughly flat plateau in the central region. This makes data far more practical to bin, display, and interpret.

Which variable gets which advantage?

- **Advantage 1** (Lorentz invariance of Δy): applies **strictly to y only**. η is purely geometric and does *not* transform additively under boosts. However, since $y \approx \eta$ at high energy, $\Delta\eta \approx \Delta y$ is approximately invariant in practice.
- **Advantage 2** (flat distributions): applies to **both y and η** in practice, since $dN/d\eta \approx dN/dy$ when $m/p_T \ll 1$ (true for most particles at colliders).

Answer (2 sentences max per sub-question)

(1) Rapidity differences Δy are *Lorentz invariant* under longitudinal boosts ($y' = y + y_F$), so distributions keep the same shape in any frame—this applies strictly to y only, but approximately to η since $\eta \approx y$ at high energy. **(2)** Distributions dN/dy and $dN/d\eta$ are approximately *flat* over a broad central region, whereas $dN/d\theta$ is extremely forward/backward peaked and impractical to use—this advantage applies to both y and η.

Part (d)

Original Question (verbatim)

Explain the reasoning behind saying that, and to what extent is it actually true, that the scale/size/"units" of either y or η should be considered as radians? Include in your answer reference to the difference in rapidity between two particles, (or equivalently, between two directions, or two reference frames).

What This Is Really Asking (Plain English)

Rapidity and pseudorapidity are dimensionless numbers, yet people say "one unit of y (or η) $\approx$ one radian." This question asks three things:

(d.1) *Where does this claim come from?* (Hint: compute $|d\eta/d\theta|$ at $\theta = 90°$.)
(d.2) *To what extent is it actually true?* Is it exact everywhere, or only at one special angle? If only at one angle, how does it extend to other rapidities?
(d.3) *Interpret Δy between two particles (or two frames) in terms of this "radians" language.* What does a rapidity difference of 1 physically correspond to?

Teaching & Derivation

From the pseudorapidity definition $\eta = -\ln[\tan(\theta/2)]$, differentiate:

$$\frac{d\eta}{d\theta} = -\frac{1}{\sin\theta}$$

At $\theta = 90°$ (i.e. $\eta = y = 0$):

$$\left|\frac{d\eta}{d\theta}\right|_{\theta=90°} = \frac{1}{\sin 90°} = 1 \qquad \Longrightarrow \qquad \boxed{|d\eta| = |d\theta| \text{ (in radians) at } \theta = 90°}$$

So at midrapidity, one unit of η equals one radian of polar angle *exactly*.

Away from $90°$**:** $|d\eta/d\theta| = 1/\sin\theta \neq 1$, so in the *lab frame* one unit of η does *not* equal one radian of θ at forward/backward angles. In fact, at small θ (large η), a single unit of η corresponds to a *tiny* amount of θ (milliradians).

But here is the key insight: Rapidity differences Δy are Lorentz invariant. A particle at large y in the lab frame is at $y = 0$ in some boosted frame (its own "rest-like" frame). In that frame, $\Delta y = 1$ corresponds to $\Delta\theta = 1\,\text{rad}$. Since Δy doesn't change under the boost, the angular meaning "1 unit of rapidity = 1 radian in the COM frame" is valid **at all** y, even though the lab-frame angle is very different.

Interpreting Δy: If two particles have $\Delta y = 1$, they are separated by $\sim$1 radian ($\sim 57°$) in the center-of-mass frame, regardless of what the lab-frame opening angle is. If two frames differ by rapidity y_F, the boost has "compressed" y_F radians of COM angular space into whatever small lab angle corresponds to that rapidity.

Answer (2 sentences max per sub-question)

At $\theta = 90°$ one can show $|d\eta/d\theta| = 1/\sin\theta = 1$, so $|d\eta| = |d\theta|$ (in radians) exactly at midrapidity; because Δy is Lorentz invariant, this "one unit = one radian in the COM frame" interpretation is carried by boost invariance to *all* values of y, even though the corresponding lab-frame $\Delta\theta$ shrinks to milliradians at forward angles. Thus a rapidity difference $\Delta y = 1$ between two particles (or two directions/frames) always corresponds to approximately 1 radian of angular separation as seen in the center-of-mass frame.

Part (e)

Original Question (verbatim)

Why is the use of (pseudo) rapidity especially advantageous for high energy particle collisions involving at least one hadron or two (X+h or h+h). Hint: consider the so-called "parton distribution functions" pdf(x) representing the probability that a parton inside a hadron carries a fraction x, which randomly fluctuates event by event, of its parent hadron's 4-momentum.

What This Is Really Asking (Plain English)

In e^+e^- collisions, using θ is fine. But in hadron collisions (pp, ep, Au+Au, ...), rapidity becomes essential. Why?

The key idea you need: in a hadron, the quarks and gluons ("partons") each carry a *random fraction* x of the hadron's total momentum. So in a pp collision, the two partons that actually scatter have momentum fractions x_1 and x_2, which are different and fluctuate randomly from one event to the next.

Explain: What does this random $x_1 \neq x_2$ do to the center-of-mass frame of the hard scattering? And why does rapidity handle this gracefully while θ does not?

Teaching & Derivation

The problem with hadron collisions: In a pp collision at the LHC, the "real" collision is between a parton from proton 1 (carrying fraction x_1) and a parton from proton 2 (carrying fraction x_2). The parton–parton center-of-mass frame is boosted relative to the lab by:

$$y_{\text{boost}} = \frac{1}{2}\ln\left(\frac{x_1}{x_2}\right)$$

Since x_1 and x_2 are drawn from broad parton distribution functions (PDFs) and vary randomly event by event, this boost is **unknown and different for every collision**.

Why θ fails: If you measure the scattering angle θ of the outgoing particles in the lab, it is a complicated convolution of the "true" partonic scattering angle *and* the random boost y_{boost}. Averaging over many events smears θ in a non-trivial, hard-to-undo way. Different events with the same underlying physics look different in θ.

Why rapidity works: Because $y' = y + y_{\text{boost}}$, the random boost just *shifts* the rapidity distribution by a constant. Crucially:

1. **Δy is invariant:** Angular separations between particles (e.g. jet cone size ΔR) are the same no matter what x_1/x_2 happened to be. You never need to reconstruct the partonic CM frame.

2. **dN/dy shape is invariant:** The distribution of produced particles in rapidity keeps its shape—the random boost just slides it left or right. Averaging over events gives a smooth plateau.

3. **Invariant cross section:** $E\,d^3\sigma/d^3p = d^3\sigma/(dy\,d^2p_T)$ is Lorentz invariant, making (y, p_T) the natural coordinate system.

Contrast with e^+e^-: In e^+e^-, the *entire* beam participates (no substructure), so $x_1 = x_2 = 1$ always. The CM frame is known and fixed $\Rightarrow$ θ works fine. It is the hadronic substructure (PDFs, random x) that makes rapidity indispensable.

Answer (2 sentences max per sub-question)

In hadron collisions, each parton carries a random momentum fraction x, so the parton–parton CM frame is boosted by $y_{\text{boost}} = \frac{1}{2}\ln(x_1/x_2)$ which fluctuates randomly event by event; since Δy is Lorentz-invariant, rapidity distributions and particle separations are unaffected by this unknown boost, unlike anything expressed in θ. This makes (pseudo)rapidity the natural variable: it absorbs the random partonic boosts into a trivial shift, so that dN/dy shapes, jet sizes (ΔR), and invariant cross sections $d^3\sigma/(dy\,d^2p_T)$ are directly comparable across events and experiments without needing to know x_1 and x_2.

Summary Table of Key Formulas

Quantity	**Class Definition**	**Textbook Definition**
Rapidity y	$\frac{1}{2}\ln\frac{E+p_z}{E-p_z} = \operatorname{arctanh}(\beta\cos\theta)$	$\operatorname{arctanh}(\beta)$
Pseudorapidity η	$-\ln\left[\tan\frac{\theta}{2}\right] = \operatorname{arctanh}\left(\frac{p_z}{\lvert\vec{p}\rvert}\right)$	—
$y = \eta$ when	$m = 0$ (exactly) or $E \gg m$ (approx.)	—
Boost transform	$y' = y + y_F$ (additive!)	Same
Δy invariance	$\Delta y' = \Delta y$ (Lorentz invariant)	Same
Radians interp.	$\lvert d\eta\rvert = \lvert d\theta\rvert$ at $\theta = 90°$; all y by inv.	—
Transverse mass	$E = m_T\cosh y,\ \ p_z = m_T\sinh y$	$E = m\cosh y,\ \ p = m\sinh y$
Partonic boost	$y_{\text{boost}} = \frac{1}{2}\ln(x_1/x_2)$	—

Advantage	**Applies to y?**	**Applies to η?**	**Applies to θ?**
Δ is Lorentz invariant	**Yes** (exactly)	∼Yes (approx.)	**No**
Flat $dN/d(\cdot)$ distributions	**Yes**	**Yes**	**No**
Units ≈ radians (in COM)	**Yes**	**Yes**	Trivially yes
Finite beam value	**Yes** (y_b finite)	**No** ($\eta_b \to \infty$)	Yes ($\theta = 0$)
No mass/PID needed	**No** (needs E)	**Yes**	**Yes**
Handles random PDF boosts	**Yes**	∼Yes	**No**